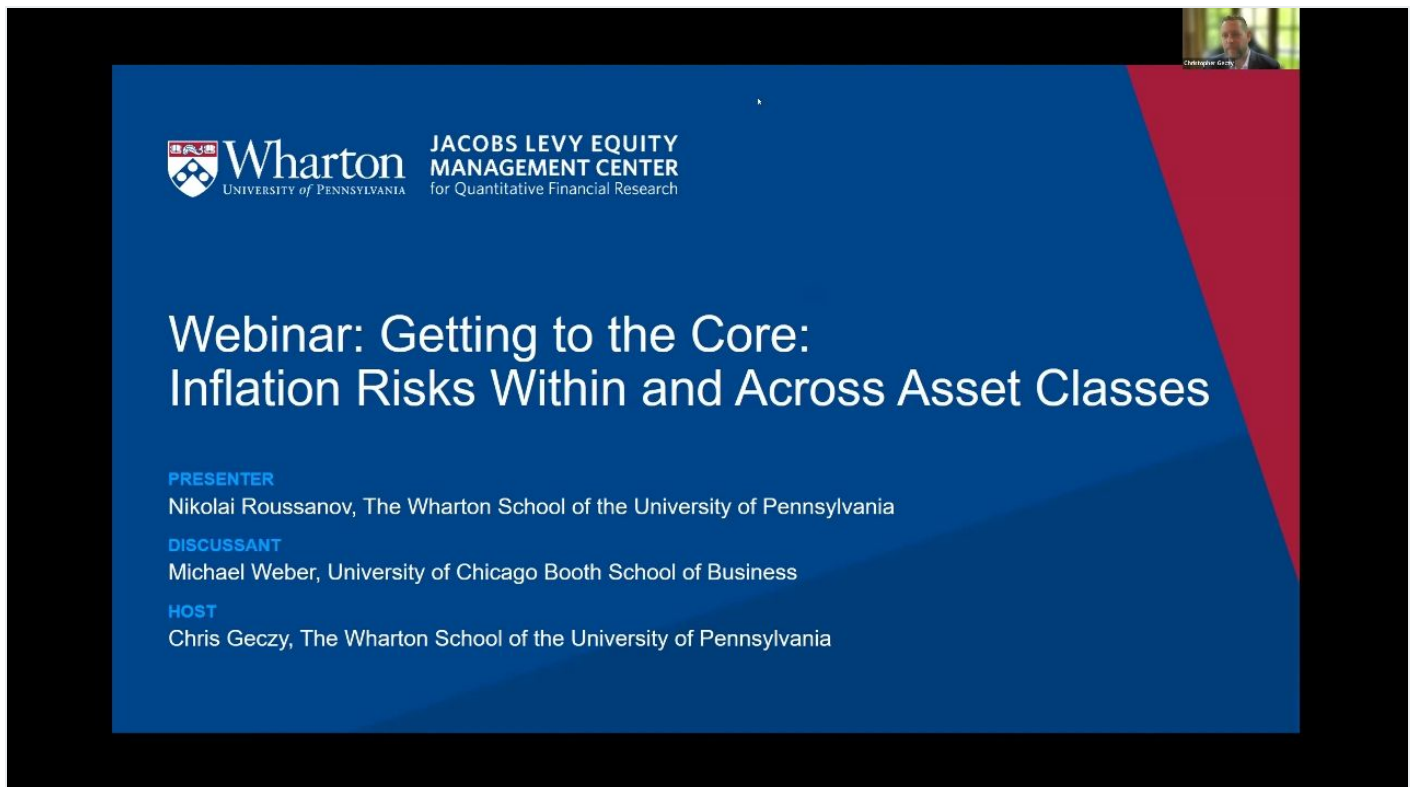


Getting to the Core Webinar



A webinar title slide with a blue background and a red diagonal stripe on the right. The Wharton University of Pennsylvania logo is in the top left, and the Jacobs Levy Equity Management Center logo is in the top right. The title is in large white text. Below the title, the presenter, discussant, and host are listed in smaller white text. A small video inset of the host is in the top right corner.

 **JACOBS LEVY EQUITY MANAGEMENT CENTER**
for Quantitative Financial Research

Webinar: Getting to the Core: Inflation Risks Within and Across Asset Classes

PRESENTER
Nikolai Roussanov, The Wharton School of the University of Pennsylvania

DISCUSSANT
Michael Weber, University of Chicago Booth School of Business

HOST
Chris Geczy, The Wharton School of the University of Pennsylvania

Getting to the Core: Inflation Risks within and across Asset Classes

Nikolai Roussanov

The Wharton School, University of Pennsylvania, and NBER

based on joint work with

Xiang Fang and Yang Liu, University of Hong Kong



GROUP

- ▶ Which assets can protect against inflation, and at what cost?
 - ▶ Conventional wisdom: currencies, commodities, and real estate are hedges, stocks are “real” assets
 - ▶ Empirically, the price of inflation risk is ambiguous

- ▶ Which assets can protect against inflation, and at what cost?
 - ▶ Conventional wisdom: currencies, commodities, and real estate are hedges, stocks are “real” assets
 - ▶ Empirically, the price of inflation risk is ambiguous
- ▶ Fama and Schwert (1977), Fama (1981), Boudoukh and Richardson (1993), Bekaert and Wang (2010), Katz, Lustig, and Nielsen (2017)

Which Inflation?

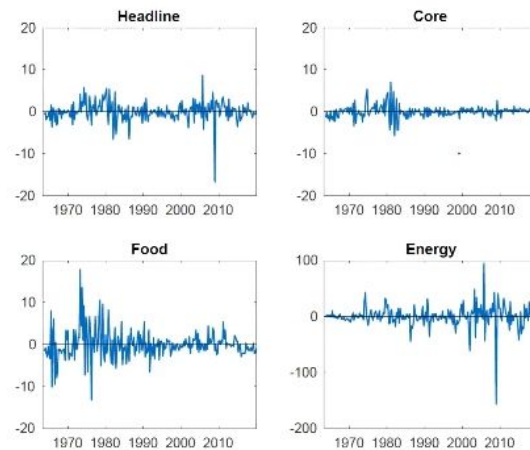


CPI Inflation and its Components

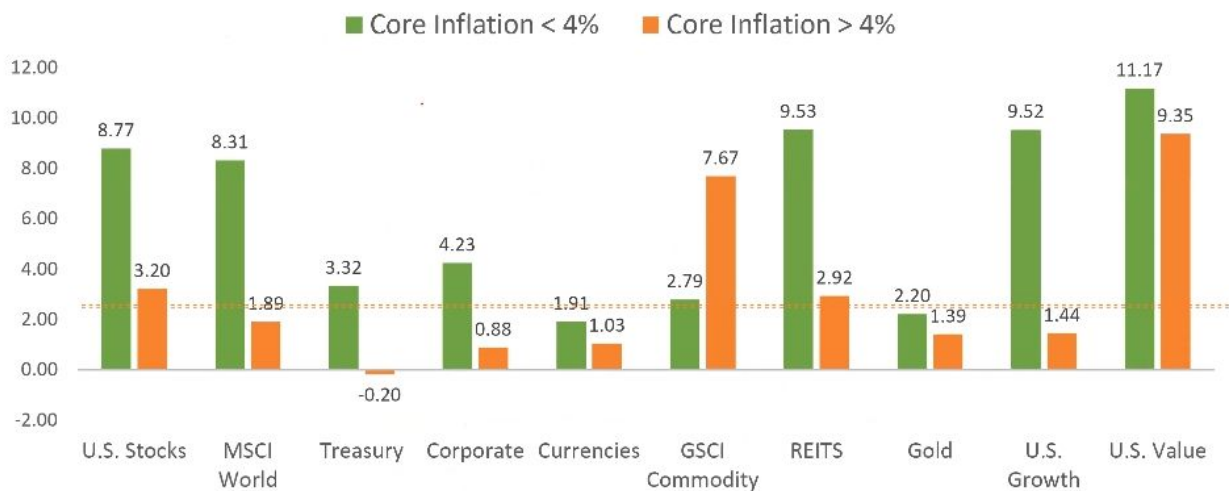
	Headline	Core	Food	Energy
A. Summary Statistics				
Mean	3.76	3.75	3.75	4.01
Std	3.24	2.66	4.04	19.52
Persist	0.60	0.79	0.43	0.04
B. Contribution to Headline				
	1.00	0.71	0.20	0.09

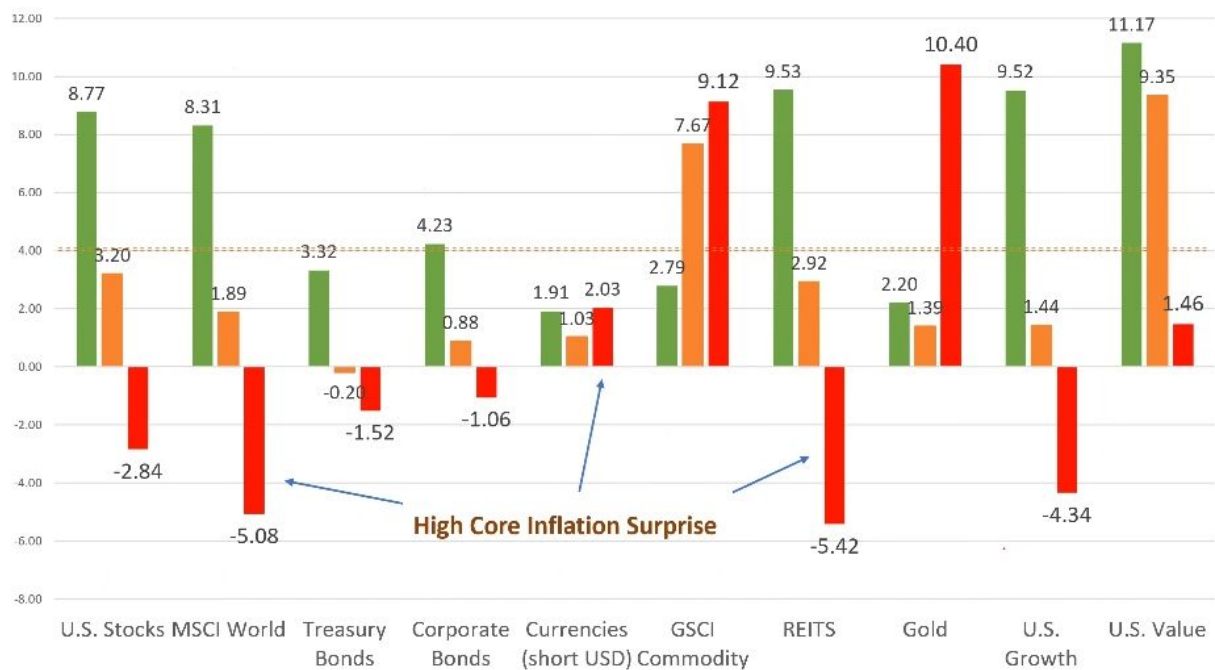
Inflation Shocks over Time

- ▶ VAR, $Y_t = c + AY_{t-1} + u_t$, u_t as shocks
- ▶ Y_t includes headline inflation and its components, p/d ratio, risk-free rate, and output gap

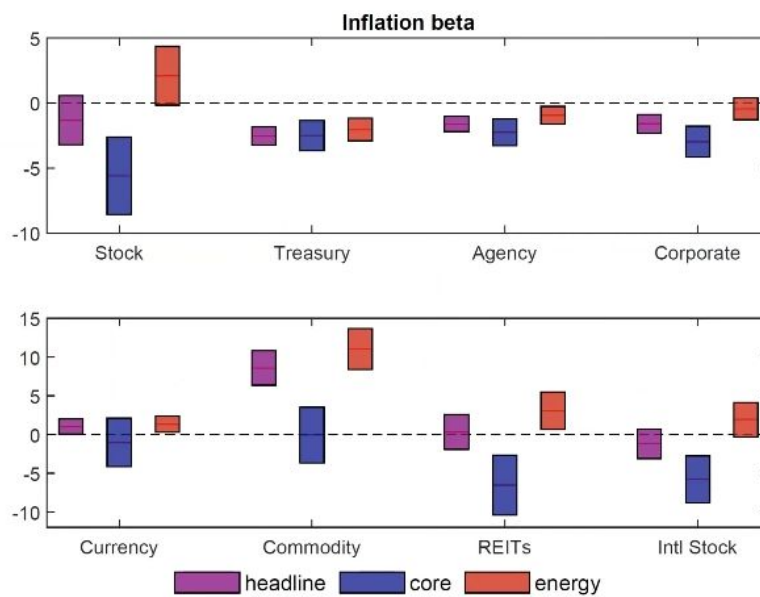


Returns when Core Inflation is Low or High

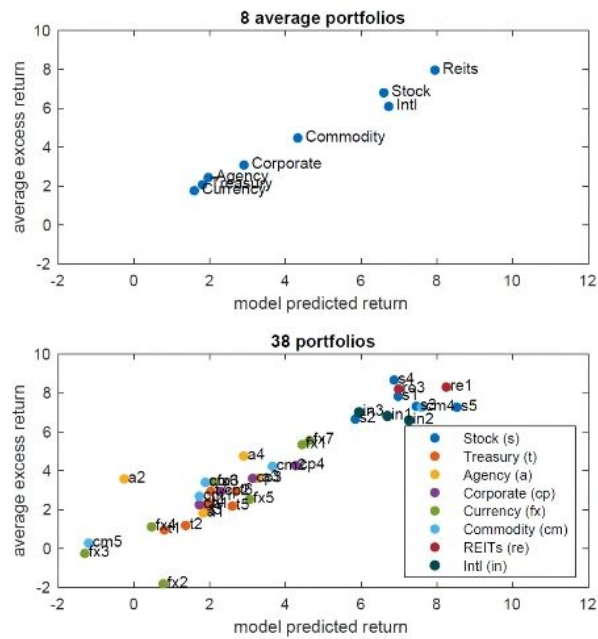




Asset Class Betas with Inflation Shocks



How Costly to Hedge? Beta vs. Average Return



Aren't Stocks "Real" Assets?

Aren't Stocks "Real" Assets?

- Fama (1981) proxy effect hypothesis: stock return-inflation relation due to inflation proxying for real variables

	headline	t-stat	R ²	core	t-stat	energy	t-stat	R ²
	1 quarter							
GDP	-0.14	(-1.21)	0.02	-0.21	(-1.88)	0.00	(-0.23)	0.03
Cons	-0.22	(-2.42)	0.08	-0.22	(-2.32)	-0.01	(-0.86)	0.07
Div	-0.27	(-1.15)	0.02	-0.67	(-4.27)	0.04	(0.96)	0.06
	1 year							
GDP	-0.75	(-2.34)	0.08	-0.70	(-2.24)	-0.05	(-1.05)	0.07
Cons	-0.66	(-2.24)	0.09	-0.46	(-1.81)	-0.05	(-1.12)	0.05
Div	-1.26	(-1.12)	0.03	-2.93	(-5.78)	0.18	(0.95)	0.11

- Core inflation negatively predicts future GDP, consumption, and dividends, especially at 1-year horizon

Cash-flow vs. Discount-rate News

- Return news can be decomposed into CF and DR news

$$(E_{t+1} - E_t)r_{t+1} = \underbrace{(E_{t+1} - E_t) \sum_{j=0}^{\infty} \rho^j \Delta d_{t+1+j}}_{N_{CF}} - \underbrace{(E_{t+1} - E_t) \sum_{j=1}^{\infty} \rho^j r_{t+1+j}}_{N_{DR}}$$

	Cash Flow News				Discount Rate News			
Mkt	Core β	t-stat	Energy β	t-stat	Core β	t-stat	Energy β	t-stat
Mkt	-2.14	(-4.12)	-0.01	(-0.23)	4.23	(3.47)	-0.19	(-2.05)
Gr	-4.96	(-5.58)	-0.11	(-1.60)	2.57	(2.57)	-0.24	(-3.14)
BM2	-2.44	(-2.83)	0.00	(-0.03)	3.07	(3.55)	-0.10	(-1.55)
BM3	-2.28	(-2.76)	-0.03	(-0.47)	2.73	(3.37)	-0.14	(-2.26)
BM4	0.71	(0.80)	0.12	(1.80)	6.27	(4.33)	-0.12	(-1.07)
VI	1.27	(1.17)	0.08	(0.92)	7.17	(4.35)	-0.14	(-1.09)

- For the stock market portfolio, negative core betas come from both CF and DR news, positive energy betas mainly come from DR news

Stock-Bond Correlation over Time

- ▶ Stock-bond correlation turned negative after 1999 (Song, 2016; Campbell et al, 2017)

	A. Headline		B. Core and Energy			
	headline	t-stat	core	t-stat	energy	t-stat
1963-1999						
Stock	-5.42	(-4.20)	-5.19	(-3.26)	-0.24	(-1.01)
Treasury	-2.88	(-5.52)	-2.77	(-4.31)	-0.20	(-2.03)
2000-2019						
Stock	2.96*	(2.22)	-6.30	(-1.18)	0.35*	(2.63)
Treasury	-2.23	(-4.73)	-0.29	(-0.15)	-0.22	(-4.65)

Note: * indicates a significant change across the two subsamples.

- ▶ Inflation and asset returns
 - ▶ First subsample: negative for stocks and bonds
 - ▶ Second subsample: positive for stock, negative for bonds
 - Driven by energy (switched signs, increased contribution)

Other “Real” Asset Classes?

- Foreign exchange
- Commodity Futures
- Real Estate

Currency Carry Trade Portfolios

		A. Headline			B. Core and energy		
	Mean	Headline β	t -stat	core β	t -stat	energy β	t -stat
		<i>Currency</i>					
Dcarry	5.34	-0.98	(-1.52)	-4.17	(-2.08)	0.00	(-0.04)
Carry-1	-1.81	0.33	(0.57)	-0.52	(-0.28)	0.06	(0.95)
Carry-2	-0.25	1.60	(2.99)	1.72	(1.03)	0.14	(2.55)
Carry-3	1.12	1.02	(1.92)	-0.04	(-0.02)	0.11	(2.02)
Carry-4	2.53	0.45	(0.74)	-2.50	(-1.34)	0.10	(1.60)
Carry-5	3.43	1.44	(2.28)	-1.28	(-0.65)	0.19	(2.94)
Carry-6	5.56	1.38	(1.87)	-3.62	(-1.60)	0.20	(2.72)

The U.S. Dollar

- USD **appreciates** when U.S. core inflation rises
- after controlling for the role of energy
- Monetary channel - Fed tightening (as they are now)
- Real Rate channel - domestic goods more valuable as RGDP falls

Commodity Futures

- Commodities: offsetting effects of price level and real growth

	Mean	A. Headline		B. Core and Energy			
		Headline β	t -stat	Core β	t -stat	Energy β	t -stat
Agriculture	0.28	4.20	(3.28)	2.06	(0.96)	0.26	(1.66)
Energy	7.26	16.51	(7.05)	-0.76	(-0.11)	1.78	(7.54)
Ind metals	4.23	4.73	(2.98)	-1.07	(-0.39)	0.66	(3.66)
Livestock	2.70	1.24	(1.24)	-1.09	(-0.66)	0.15	(1.22)
Pre metals	3.41	3.28	(2.65)	-0.22	(-0.11)	0.43	(2.96)
Gold	1.98	2.14	(1.97)	1.74	(0.91)	0.24	(1.92)
Silver	3.52	4.95	(2.63)	-0.09	(-0.03)	0.68	(3.06)
Platinum	4.36	3.40	(2.29)	7.51	(1.63)	0.26	(1.69)

Concluding

- ▶ Shed new light on the nature of inflation risk: core and energy
 - ▶ Conventional inflation "hedges" only protect against energy inflation, not the core inflation
 - ▶ Core inflation carries a negative risk premium, consistently estimated within and across asset classes
- ▶ New insights into the changing stock-bond correlation

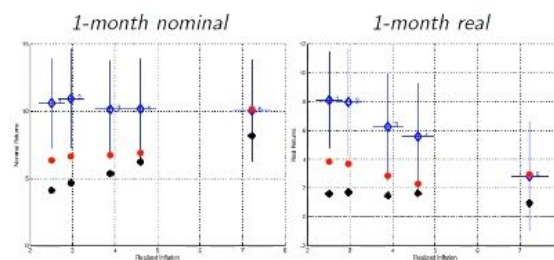
Getting to the Core: Inflation Risks Within and Across Asset Classes

Xiang Fangy, Yang Liuy, and Nikolai Roussanov

discussed by Michael Weber

May 11, 2022

Incomplete Pass Through of Inflation into Nominal Returns

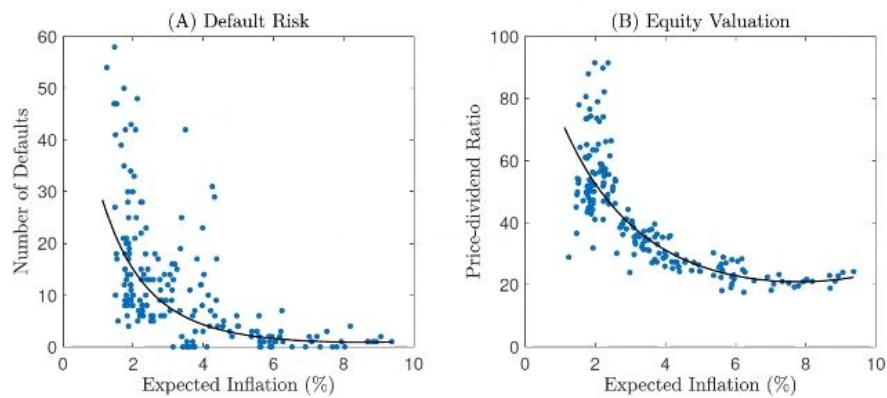


Notes: Portfolios of countries sorted by lagged inflation. We plot stock returns ('diamonds'), bond returns ('circles'), and returns on T-bills ('stars').

Source: Katz, Lustig, Nielsen (RFS, 2017)

- Incomplete pass through of local inflation into nominal returns
- Consistent w sticky discount rates or nominal cash flow extrapolation

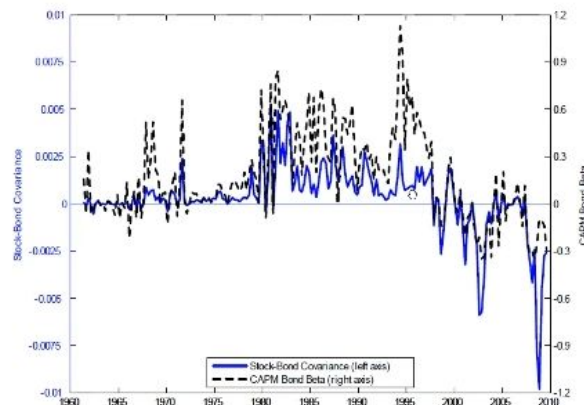
High Expected Inflation, Default, and Valuation



Source: Bharma, Dorion, Jeanneret, and Weber (RFS, 2022)

- High expected inflation periods of low default and low equity valuation
- Due to sticky cash flows and leverage

Changing Sign in the Stock-Bond Covariance



Source: Campbell, Sunderam, and Viceira (2015)

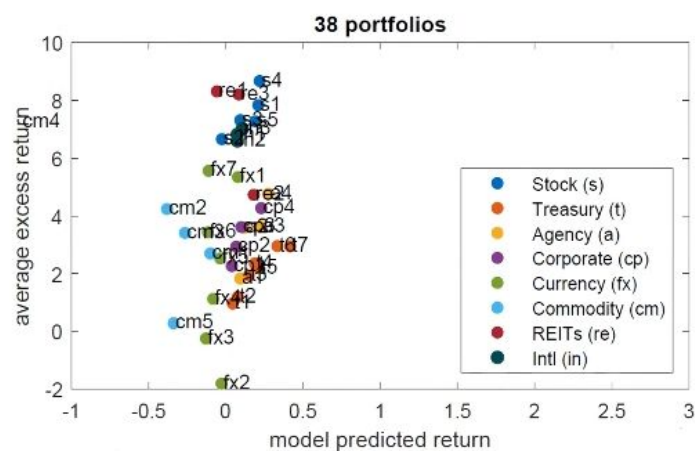
- Changing sign in the stock-bond covariance
- Stocks and bonds positively correlated in the pre 2000 period
- Coincides with change in conduct of monetary policy
- Also change in cyclical properties of inflation process

What are Inflation Hedging Properties of Real Assets?

Boons, Duarte, de Roon, and Szymanowska (JFE, 2020)

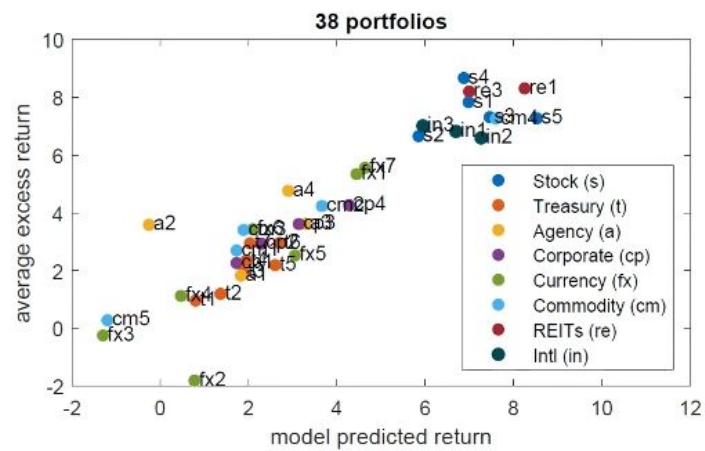
- Inflation risk is priced in stock returns
- Small unconditional price of inflation risk
- Inflation risk premia in XS and TS vary over time
- Inflation risk premia change sign in early 2000
- Time variation is due to both price and quantities of risk
- Inflation priced because inflation predicts real consumption growth
- Historical changes in predictability explain time variation

Headline Inflation Betas and the XS



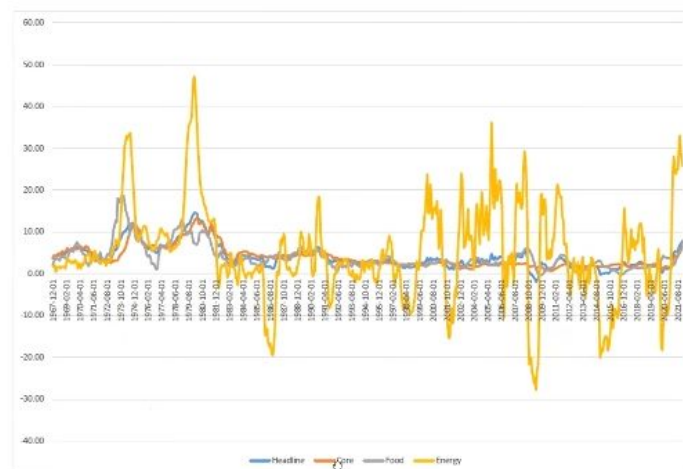
- Overall inflation betas unrelated to spread in returns

Core Inflation Betas and the XS



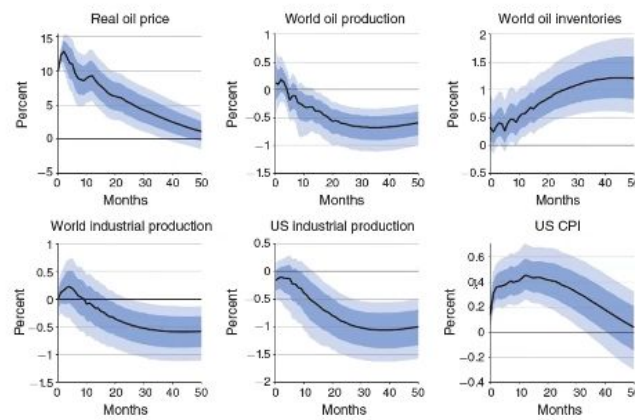
- Core inflation betas line up with differences in average returns

Different Cyclical Properties of Inflation



- Energy (and food) inflation volatile and quickly mean reverting
- Change in cyclical properties of overall inflation over time
- Low interest rate environment since 2008

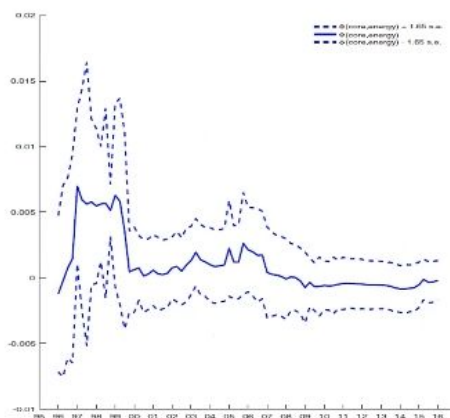
Pass Through of Energy Inflation into Headline Inflation



Source: Kaenzig (AER, 2021)

- Oil news shock increases US headline inflation

Pass Through of Energy Inflation into Core Inflation



Source: Ajello, Benzoni, and Chyruk (RFS, 2020)

- Zero pass through of enery inflation into core inflation

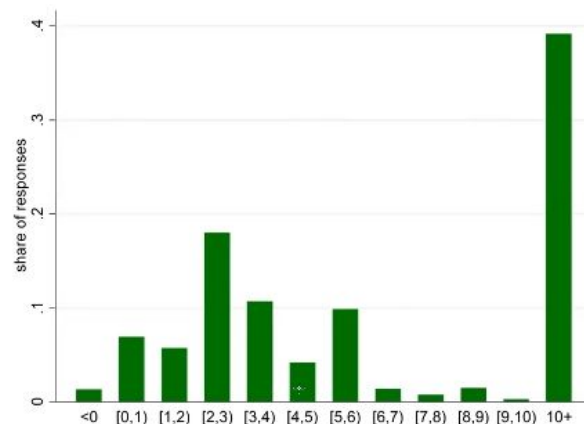
Inflation and Monetary Policy

- Non-core inflation highly mean reverting
- Little predictive power for future inflationary pressure
- Monetary policy: focus on core inflation for inflationary pressures
- Higher core inflation → higher policy rates
- “Of course, if commodity and energy prices were to lead to a general expectation of a broader increase in inflation, more substantial policy rate increases would be justified. But assuming there is a generally high degree of centralbank credibility, there is no reason for such expectations to develop”

Charly Evans, President and CEO of Chicago Fed

BUT Households and Firms Lack Basic knowledge...

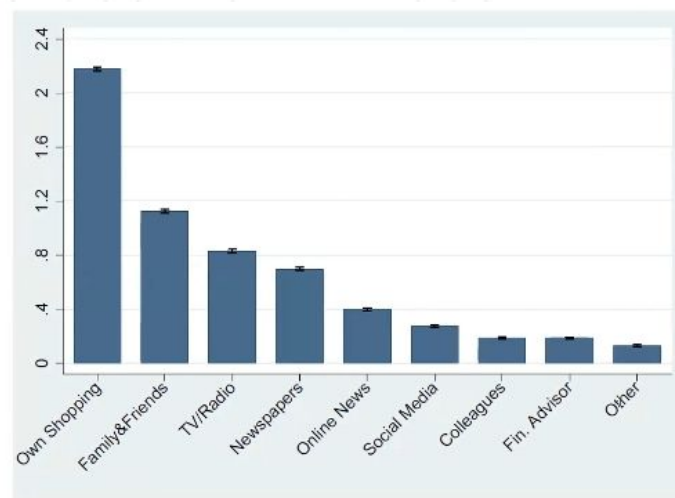
Inflation Target of the Fed in 2018 as perceived by 25,000 Americans



Source: Coibion, Gorodnichenko, Weber, JPE (2022):

- Expectations react to info and determine consumption choices

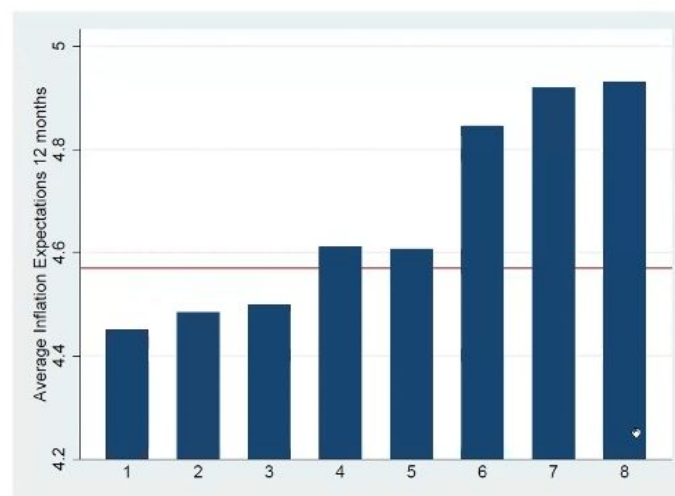
But How Do Agents Form Expectations?



Source: D'Acunto, Malmendier, Ospina, and Weber (JPE, 2021)

- Shopping most important source of information for future overall inflation

But How Do Agents Form Expectations?



Source: D'Acunto, Malmendier, Ospina, and Weber (JPE, 2021)

- Variation in shopping bundle inflation → heterogeneity in overall inflation expectations

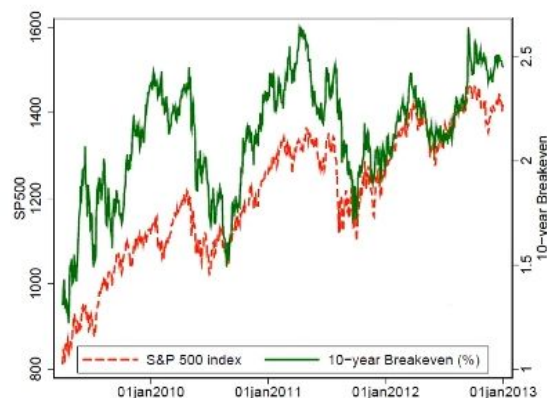
Monetary Policy Response as Driver

- Core inflation betas systematically negative
- Because of more than one for one response of monetary policy?
- Higher core inflation triggers increase in real interest rates
- Future consumption growth should be lower
- Authors run event study of inflation news on FFF and stock futures
- Stock futures have negative reaction controlling for FFF
- But empirically, tighter monetary policy also increases risk premia

Bernanke and Kuttner (JF, 2005)

- Negative core betas might still be reflection of policy reaction

Is Higher Inflation Always Bad?



Source: Gourio and Ngo (2020)

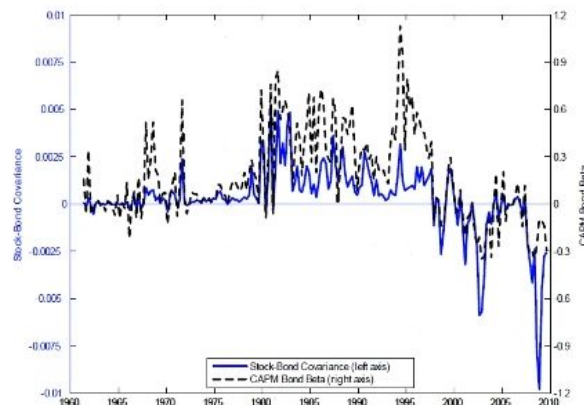
- Higher inflation expectations can comove positively with higher stock valuations
- During binding ZLB, higher inflation does not trigger increase in policy rates
- Higher inflation expectations hence lower real interest rates (Fisher equation)
- Lower real interest rates stimulate aggregate demand (Euler equation)

Subsample Analysis

	A. Headline		B. Core and Energy			
	headline β	t -stat	core β	t -stat	energy β	t -stat
1963-1999						
Stock	-5.42	(-4.20)	-5.19	(-3.26)	-0.24	(-1.01)
Treasury	-2.88	(-5.52)	-2.77	(-4.31)	-0.20	(-2.03)
Agency	-2.49	(-4.31)	-2.50	(-3.56)	-0.12	(-1.20)
Corporate	-3.23	(-5.76)	-3.08	(-4.44)	-0.22	(-2.22)
Currency	0.12	(0.08)	0.26	(0.09)	0.06	(0.33)
Commodity	4.55	(2.80)	0.42	(0.21)	0.65	(2.24)
REIT	-5.79	(-3.76)	-5.91	(-3.29)	-0.37	(-1.43)
Intl Stock	-5.92	(-4.60)	-5.12	(-3.24)	-0.46	(-1.97)
2000-2019						
Stock	2.96	(2.22)	-6.30	(-1.18)	0.35	(2.63)
Treasury	-2.23	(-4.73)	-0.29	(-0.15)	-0.22	(-4.65)
Agency	-1.05	(-4.38)	-0.27	(-0.27)	-0.10	(-4.11)
Corporate	-0.20	(-0.49)	-0.66	(-0.40)	-0.01	(-0.22)
Currency	1.30	(2.67)	-1.87	(-0.96)	0.15	(3.10)
Commodity	12.44	(8.35)	-0.38	(-0.06)	1.24	(8.44)
REIT	3.47	(2.22)	-10.18	(-1.65)	0.44	(2.88)
Intl Stock	3.28	(2.47)	-6.62	(-1.25)	0.39	(2.95)

- Headline betas change sign over time
- Energy betas also flip signs
- Core betas consistently negative

Changing Sign in the Stock-Bond Covariance

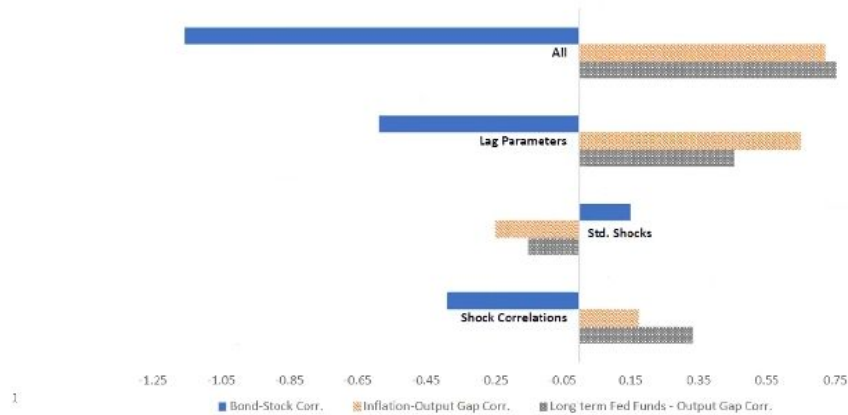


¹ Source: Campbell, Sunderam, and Viceira (2015)

- Changing sign in the stock-bond covariance
- Stocks and bonds positively correlated in the pre 2000 period
- Coincides with change in conduct of monetary policy
- Also change in cyclical properties of inflation process

Monetary Policy as Driver

Figure 6: Counterfactuals



Source: Campbell, Pflueger, and Viceira (JPE, 2020)

- Changing parameters in policy rule biggest driver of change in stock-bond correlation

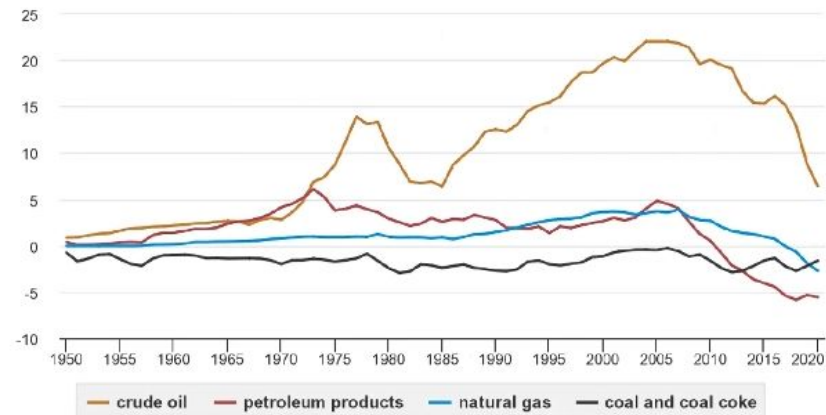
Drivers of Inflation

- Energy inflation often due to supply shocks such as wars, etc
- Hence, energy inflation bad for US?
- Depends on whether US is net exporter or importer of energy

Net Energy Imports

U.S. energy net imports by major source, 1950-2020

quadrillion British thermal units



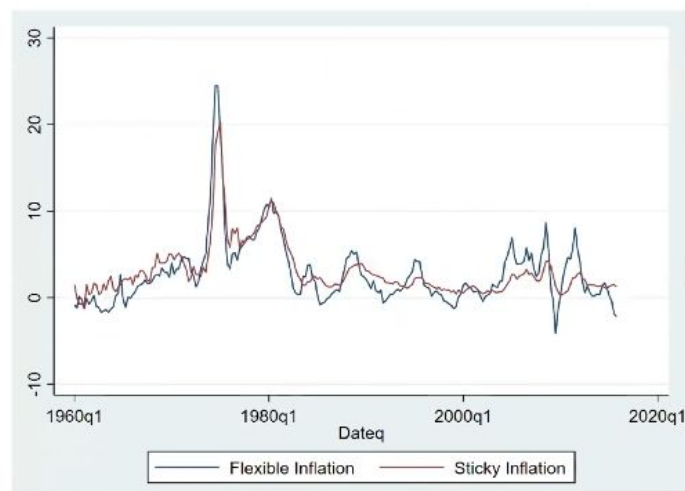
Source: U.S. Energy Information Administration, *Monthly Energy Review*, Table 1.4 c, April 2021, preliminary data for 2020

- Large decrease in net imports of crude oil since 2000

Core versus Energy Inflation

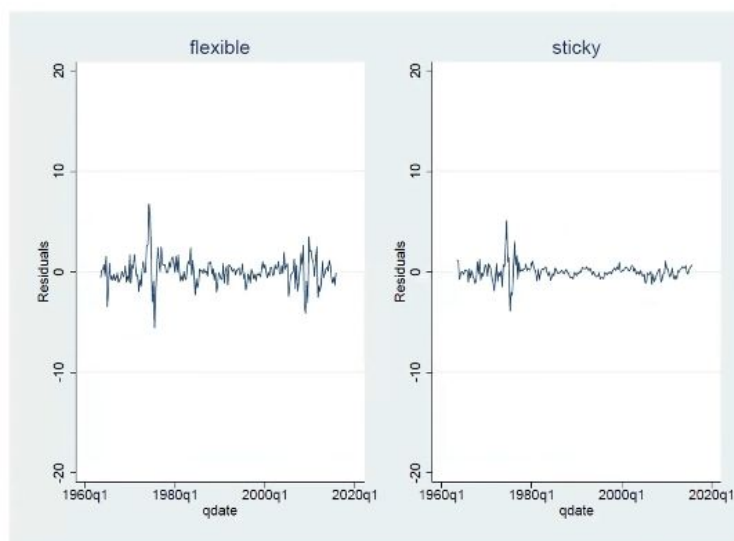
- Authors conjecture price stickiness as fundamental distinguishing force
- Use frequencies of price adjustments from data underlying PPI
Gorodnichenko and Weber (AER, 2016)
- Merge frequencies with sectoral PPI inflation at 6-digit NAICS level
- Calculate sticky and flexible inflation based on median frequency

Sticky and Flexible Sector Inflation



- Flexible inflation more volatile consistent with energy inflation

Sticky and Flexible Shock Series



- Flexible inflation series highly correlated with energy inflation (64%)
- Sticky inflation series highly correlated with core inflation (86%)

Model: Price of Energy Inflation

- Price of risk for energy inflation ambiguous
 - Affected by both demand and supply shocks
- pre-1999: energy beta of stocks negative → supply shocks important
- Demand channel driving force in first decades of the 21st century
- Empirically aim to disentangle
- More generally, what are reduced form shocks in VAR capturing

Random Rants

- The inflation innovation are generated regressors
- Alternative state variables in VAR
- Expanding window
- Simple differences
- Multivariate regressions also including other factors

Conclusion

- Great paper every one should read!
- Convincing new evidence on inflation hedging properties
- Few minor quibbles to address possibly in follow-up work